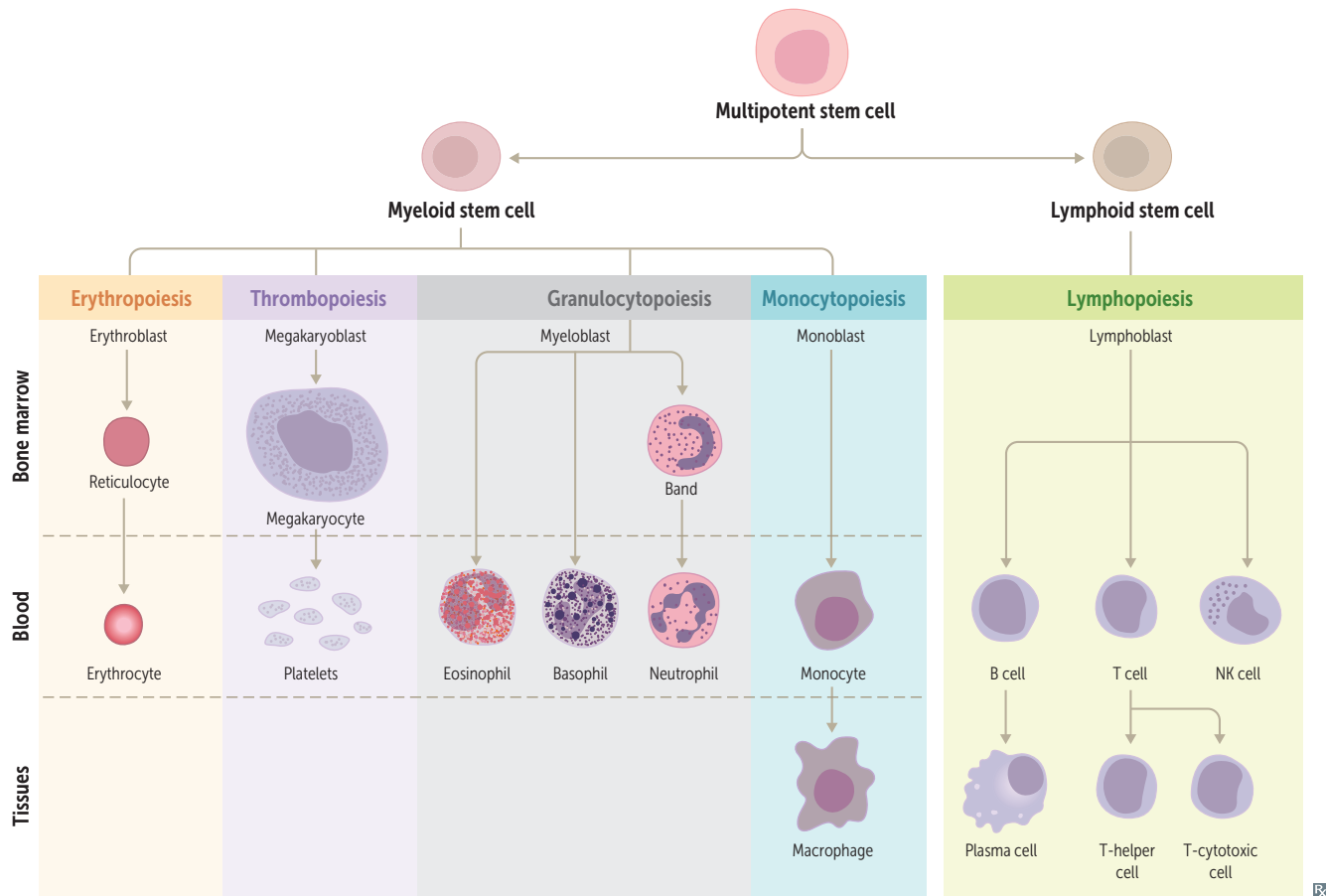
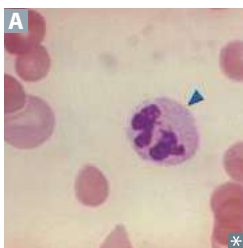


► HEMATOLOGY AND ONCOLOGY—ANATOMY

Hematopoiesis

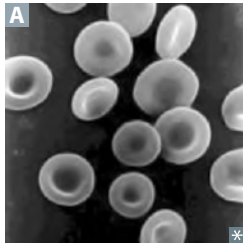


Neutrophils



Acute inflammatory response cells. Phagocytic. Multilobed nucleus **A**. Specific granules contain leukocyte alkaline phosphatase (LAP), collagenase, lysozyme, and lactoferrin. Azurophilic granules (lysosomes) contain proteinases, acid phosphatase, myeloperoxidase, and β -glucuronidase. Inflammatory states (eg, bacterial infection) cause neutrophilia and changes in neutrophil morphology, such as left shift, toxic granulation (dark blue, coarse granules), Döhle bodies (light blue, peripheral inclusions, arrow in **B**), and cytoplasmic vacuoles.

Neutrophil chemotactic agents: C5a, IL-8, LTB₄, 5-HETE (leukotriene precursor), kallikrein, platelet-activating factor, N-formylmethionine (bacterial proteins). Hypersegmented neutrophils (nucleus has 5–6+ lobes) are seen in vitamin B₁₂/folate deficiency. **Left shift**—↑ neutrophil precursors (eg, band cells, metamyelocytes) in peripheral blood. Reflects states of ↑ myeloid proliferation (eg, inflammation, CML). **Leukoerythroblastic reaction**—left shift accompanied by immature RBCs. Suggests bone marrow infiltration (eg, myelofibrosis, metastasis).

Erythrocytes

Carry O_2 to tissues and CO_2 to lungs. Anucleate and lack organelles; biconcave **A**, with large surface area-to-volume ratio for rapid gas exchange. Life span of ~120 days in healthy adults; 60–90 days in neonates. Source of energy is glucose (90% used in glycolysis, 10% used in HMP shunt). Membranes contain Cl^-/HCO_3^- antiporter, which allow RBCs to export HCO_3^- and transport CO_2 from the periphery to the lungs for elimination.

Erythro = red; *cyte* = cell.

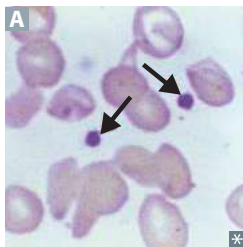
Erythrocytosis = polycythemia = ↑ Hct.

Anisocytosis = varying sizes.

Poikilocytosis = varying shapes.

Reticulocyte = immature RBC; reflects erythroid proliferation.

Bluish color (polychromasia) on Wright-Giemsa stain of reticulocytes represents residual ribosomal RNA.

Thrombocytes (platelets)

Involved in 1° hemostasis. Anucleate, small cytoplasmic fragments **A** derived from megakaryocytes. Life span of 8–10 days (pl8lets). When activated by endothelial injury, aggregate with other platelets and interact with fibrinogen to form platelet plug. Contain dense granules (Ca^{2+} , ADP, Serotonin, Histamine; **CASH**) and α granules (vWF, fibrinogen, fibronectin, platelet factor 4). Approximately 1/3 of platelet pool is stored in the spleen.

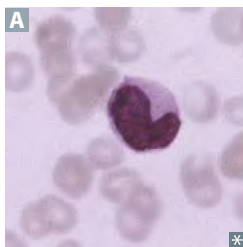
Thrombocytopenia or ↓ platelet function results in petechiae.

vWF receptor: GpIb.

Fibrinogen receptor: GpIIb/IIIa.

Thrombopoietin stimulates megakaryocyte proliferation.

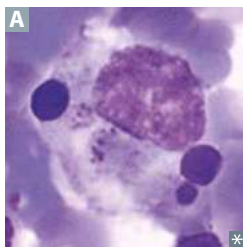
Alfa granules contain vWF, fibrinogen, fibronectin, platelet factor four.

Monocytes

Found in blood, differentiate into macrophages or dendritic cells in tissues.

Large, kidney-shaped nucleus **A**. Extensive “frosted glass” cytoplasm.

Mono = one (nucleus); *cyte* = cell.

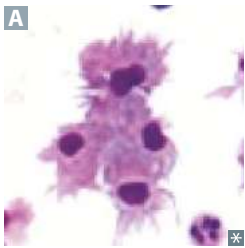
Macrophages

A type of antigen-presenting cell. Phagocytose bacteria, cellular debris, and senescent RBCs. Long life in tissues. Differentiate from circulating blood monocytes **A**. Activated by $IFN-\gamma$. Can function as antigen-presenting cell via MHC II. Also engage in antibody-dependent cellular cytotoxicity. Important cellular component of granulomas (eg, TB, sarcoidosis), where they may fuse to form giant cells.

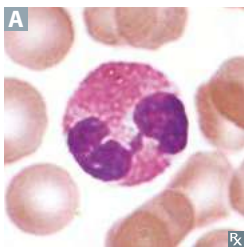
Macro = large; *phage* = eater.

Macrophage naming varies by specific tissue type (eg, Kupffer cells in liver, histiocytes in connective tissue, osteoclasts in bone, microglial cells in brain).

Lipid A from bacterial LPS binds CD14 on macrophages to initiate septic shock.

Dendritic cells

Highly phagocytic antigen-presenting cells (APCs) **A**. Function as link between innate and adaptive immune systems (eg, via T-cell stimulation). Express MHC class II and Fc receptors on surface. Can present exogenous antigens on MHC class I (cross-presentation).

Eosinophils

Defend against helminthic infections (major basic protein). Bilobate nucleus. Packed with large eosinophilic granules of uniform size **A**. Highly phagocytic for antigen-antibody complexes.

Produce histaminase, major basic protein (MBP, a helminthotoxin), eosinophil cationic protein, eosinophil-derived neurotoxin, and IL-5, which promotes eosinophilic activation and proliferation.

Eosin = pink dye; *philic* = loving.

Causes of eosinophilia (**PACMAN Eats**):

Parasites

Asthma

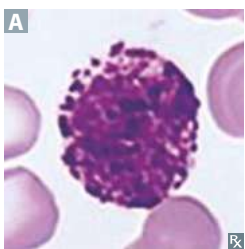
Chronic adrenal insufficiency

Myeloproliferative disorders

Allergic processes

Neoplasia (eg, Hodgkin lymphoma)

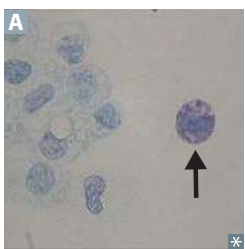
Eosinophilic granulomatosis with polyangiitis

Basophils

Mediate allergic reaction. Densely basophilic granules **A** contain heparin (anticoagulant) and histamine (vasodilator). Leukotrienes synthesized and released on demand.

Basophilic—stains readily with **basic** stains.

Basophilia is uncommon, but can be a sign of myeloproliferative disorders, particularly CML.

Mast cells

Mediate local tissue allergic reactions. Contain basophilic granules **A**. Originate from same precursor as basophils but are not the same cell type. Can bind the Fc portion of IgE to membrane. Activated by tissue trauma, C3a and C5a, surface IgE cross-linking by antigen (IgE receptor aggregation) → degranulation → release of histamine, heparin, tryptase, and eosinophil chemotactic factors.

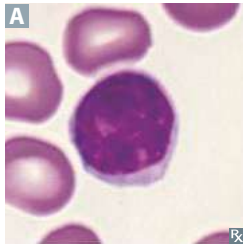
Involved in type I hypersensitivity reactions.

Cromolyn sodium prevents mast cell degranulation (used for asthma prophylaxis).

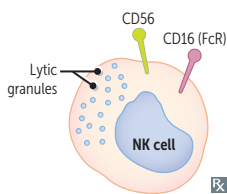
Vancomycin, opioids, and radiocontrast dye can elicit IgE-independent mast cell degranulation.

Mastocytosis—rare; proliferation of mast cells in skin and/or extracutaneous organs. Associated with *c-KIT* mutations and ↑ serum tryptase.

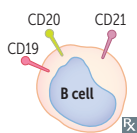
↑ histamine → flushing, pruritus, hypotension, abdominal pain, diarrhea, peptic ulcer disease.

Lymphocytes

Refer to B cells, T cells, and natural killer (NK) cells. B cells and T cells mediate adaptive immunity. NK cells are part of the innate immune response. Round, densely staining nucleus with small amount of pale cytoplasm **A**.

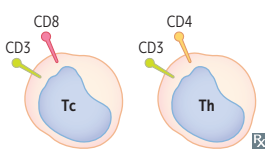
Natural killer cells

Important in innate immunity, especially against intracellular pathogens. NK cells are larger than B and T cells, with distinctive cytoplasmic lytic granules (containing perforin and granzymes) that, when released, act on target cells to induce apoptosis. Distinguish between healthy and infected cells by identifying cell surface proteins (induced by stress, malignant transformation, or microbial infections). Induce **apoptosis** (natural **killer**) in cells that do not express class I MHC cell surface molecules, eg, virally infected cells in which these molecules are downregulated.

B cells

Mediate humoral immune response. Originate from stem cells in bone marrow and matures in marrow. Migrate to peripheral lymphoid tissue (follicles of lymph nodes, white pulp of spleen, unencapsulated lymphoid tissue). When antigen is encountered, B cells differentiate into plasma cells (which produce antibodies) and memory cells. Can function as an APC.

B = bone marrow.

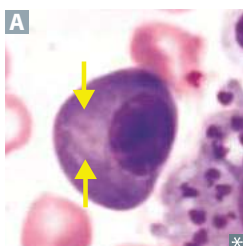
T cells

Mediate cellular immune response. Originate from stem cells in the bone marrow, but mature in the thymus. Differentiate into cytotoxic T cells (express CD8, recognize MHC I), helper T cells (express CD4, recognize MHC II), and regulatory T cells. CD28 (costimulatory signal) necessary for T-cell activation. Most circulating lymphocytes are T cells (80%).

T = thymus.

CD4+ helper T cells are the primary target of HIV.

Rule of 8: MHC **II** × CD**4** = **8**;
MHC **I** × CD**8** = **8**.

Plasma cells

Produce large amounts of antibody specific to a particular antigen. “Clock-face” chromatin distribution and eccentric nucleus, abundant RER, and well-developed Golgi apparatus (arrows in **A**). Found in bone marrow and normally do not circulate in peripheral blood.

Multiple myeloma is a plasma cell dyscrasia.